Daffodil International University

Department of Software Engineering

Quiz 2

Course Title: Software Quality Assurance & Testing		**Course Code:** SE 312
Time: 45 Minutes	**Marks:** 15	**Date:**
Student Name:	**Student ID:**	**Section:**

Answer all the questions. Marks are shown in the right margin.

1. A software team tested an application and recorded the following data. **[5]**

Size of application	Defects detected	Defects fixed	Defects rejected	Days to solve
8,000 LOC	56	49	7 (5 + 2)	14 business days

Calculate the following testing metrics. Show the formula and the steps.

(a) Defect Density **[1]** **(b)** Defect Removal Rate **[1.5]** **(c)** Defect Rejection Rate **[1.5]** **(d)** Turn Around Time **[1]**

2. Functional testing and non-functional testing check different aspects of a software application.

(a) Define functional testing and non-functional testing, and give one example of each. **[2]**

(b) Classify each scenario as Functional or Non-Functional and justify in one line. **(i)** Verify that a user can reset the password using the link sent to the registered email. **(ii)** Check that the student dashboard loads within 3 seconds. **(iii)** Check that 2,000 users can log in to the portal at the same time. **(iv)** Verify that an error message appears when an invalid card number is entered. **[3]**

3. A tester finds a defect in a software product. The defect then goes through different stages until it is finally closed. **[5]**

(a) Draw the complete Defect (Bug) Life Cycle diagram. Label every state and show the direction of all transitions. **[3]**

(b) A developer decides not to fix one defect in the current release because it is low priority, and declares another defect invalid. Name these two states and write the three possible causes of the invalid defect. **[2]**